OPD-Evolver: Cultivating Holistic Agent Evolver via On-Policy Distillation

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HarnessRSI 63 ★★

evidence: retained

Paper

Code

1 Two loops, two retained artifacts

FAST · fixed weights at evaluation
Empty repository → sequential tasks
Trajectories · tips · skills · tools

▼

Select → execute → observe feedback
Retrieve candidates; policy selects context
Environment returns success / failure

▼

Write and maintain persistent memory
Policy writes, merges and deletes memories
Maintenance every 30 tasks; not a dev gate

▼

SLOW · train policy from hindsight
Score selected vs unselected candidates
Teacher sees values + successful experience

▼

Distill on the student's own prefixes
Same-policy teacher; stop-gradient token KL
Train selection, execution, writing, upkeep

Deploy student → build a new online memory
Slow loop changes weights; fast loop retains experience, not a rewritten optimizer program.

§3; Algorithm 1. Teacher filter: score ≥ 0.01; up to 20 of 50 candidates. Writer-side: top 30%. The online placement of that filter is not fully specified.

2 Same-model memory comparison

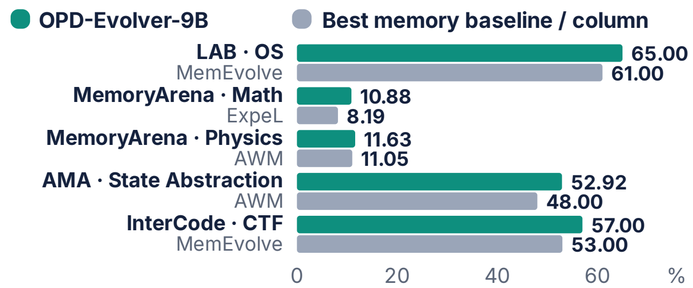


Table 1 · selected subsets · Qwen3.5-9B · task-success %. Paper reports leading all ten subsets for both sizes. No run/seed count, SD or CI is given.

3 Controls, and what is still missing

4B · Table 3 (%)	Bash	CTF	SQL
Full	36.16	34.00	45.86
− Slow loop	32.14	28.00	39.17
− Attribution	31.20	26.69	38.50
− Selection	35.27	32.00	42.04
− Writing distill.	34.38	29.00	41.08
− Maintenance	35.30	30.10	43.51

No variance / run count. − Attribution: frequency score; − Selection: similarity top-5; − Maintenance: no update training.

- **Diagnostic:** Figure 4: memory-free execution improves for both sizes; not the full memory loop.
- **Access:** memory baselines start empty; same feedback, no answer keys. OPD hindsight training is not documented for each.
- **Table 2:** same MiniHack data. Skill0 gets skills + curriculum; MemRL updates memory; Complementary RL trains actor + memory actor. No matched total compute.
- **Splits:** main sources benchmark-disjoint; Table 2 adds MiniHack training. Online memory retains earlier test lessons. No content / near-duplicate leakage audit.
- **Tuning:** selection split and baseline search budgets unreported. OPD learning rate, batch and training steps/epochs missing.
- **Cost:** hardware only; no time, tokens, dollars or GPU-hours.

4 Where the gain is thin or absent

- **Table 1, 9B:** Physics 11.63% vs AWM 11.05%; Causal Inference 47.32% vs Memp 45.81%.
- **Table 2 · MiniHack:** 51 tasks/environment. Room 98.04% vs GRPO 100.00%; Maze 27.45% vs Skill0 25.49%: one-task margins‡ if single-pass; run protocol unspecified.
- **KeyRoom:** 9.80% vs Complementary RL 5.16%. Its MiniHack row does not fit integer successes / 51; averaging unspecified.
- **Qwen3.5-397B-A17B** leads Table 1 DB, CI, SU and Bash.

‡ Conditional counts, not reported outcomes (SOURCE.md). Table 2 adds MiniHack training; no per-row backbone labels.

Takeaway

Training the memory lifecycle helps beyond simply attaching a repository. The evidence supports broad streaming-task gains, with smaller margins on some subsets — not a cost-matched, variance-tested proof of a better improver.

Our read · HarnessRSI

Score 63 · ★★ · evidence: retained

Reports: cross-env tests, ablations, empty-start memory.
Missing: seed/order uncertainty, matched compute, cost, tested leakage audit.